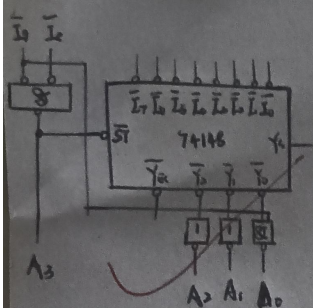


202315501 王克刚

3-2



3-4

1) $F_1(A, B, C) = AB + C$

$$= ABC + \bar{C} + CCA + \bar{C}CB$$

$$= ABC + \bar{C} + CAB + \bar{C}CB$$

$$= ABC + \bar{C} + \bar{A}BC + \bar{C}CB$$

$$= m_7 + m_6 + m_5 + m_1$$

$$= \bar{m}_7 \bar{m}_6 \bar{m}_5 \bar{m}_1$$

$$F_1(A, B, C) = AB + C$$

$$= ABC + \bar{C} + CCA + \bar{C}CB$$

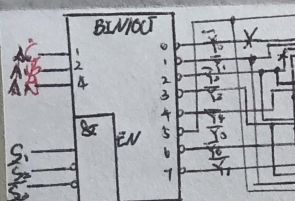
$$= ABC + \bar{C} + (CA + \bar{C}B) \bar{C}$$

$$= ABC + \bar{C} + \bar{C}B + \bar{C}C$$

$$= ABC + \bar{C} + \bar{C}B + \bar{C}C$$

$$= m_7 + m_6 + m_5$$

$$= m_7 + m_6 + m_5$$



$$F_3(A, B, C) = (\bar{C} + \bar{B})C$$

$$= \bar{C}C + \bar{B}C$$

$$= \bar{C}C + \bar{B}C$$

$$= \bar{A}BC + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$$

$$= \bar{A}BC + \bar{A}\bar{B}C + \bar{A}BC$$

$$= m_4 + m_6 + m_2 = m_5 + m_1 + m_3$$

$$= \bar{m}_5 \bar{m}_1 \bar{m}_3$$

(2) $F_7(A, B, C) = \prod M(0, 1, 4, 5)$

$$= m_2 + m_3 + m_6 + m_7$$

$$= \bar{m}_2 \bar{m}_3 \bar{m}_6 \bar{m}_7$$

3-6 标准式与表达式

$$Y_0 = \bar{A}_3 \bar{A}_2 \bar{A}_1 \bar{A}_0 = A_3 + A_2 + A_1 + A_0$$

$$Y_1 = A_0 + \bar{A}_1 + \bar{A}_2 + \bar{A}_3 = \bar{A}_0 + A_1 + A_2 + A_3$$

$$Y_2 = \bar{A}_0 + A_1 + \bar{A}_2 + \bar{A}_3 = A_0 + \bar{A}_1 + A_2 + A_3$$

$$Y_3 = A_0 + A_1 + \bar{A}_2 + \bar{A}_3 = \bar{A}_0 + \bar{A}_1 + A_2 + A_3$$

$$Y_4 = \bar{A}_0 + \bar{A}_1 + A_2 + \bar{A}_3 = A_0 + A_1 + \bar{A}_2 + A_3$$

$$Y_5 = A_0 + \bar{A}_1 + A_2 + \bar{A}_3 = \bar{A}_0 + A_1 + \bar{A}_2 + A_3$$

$$Y_6 = \bar{A}_0 + A_1 + A_2 + \bar{A}_3 = A_0 + \bar{A}_1 + \bar{A}_2 + A_3$$

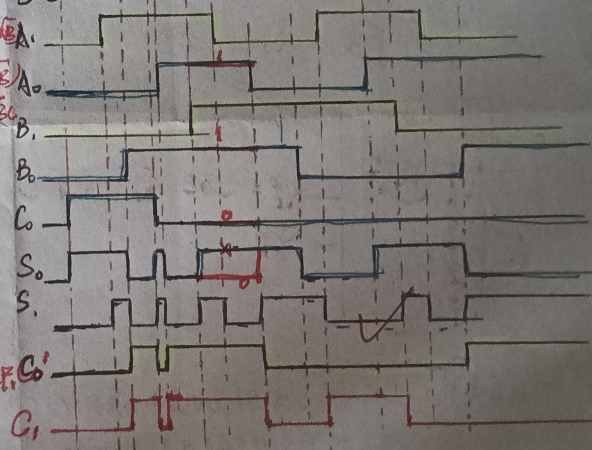
$$Y_7 = A_0 + A_1 + A_2 + \bar{A}_3 = \bar{A}_0 + \bar{A}_1 + \bar{A}_2 + A_3$$

$$Y_8 = \bar{A}_0 + \bar{A}_1 + \bar{A}_2 + A_3 = A_0 + A_1 + A_2 + \bar{A}_3$$

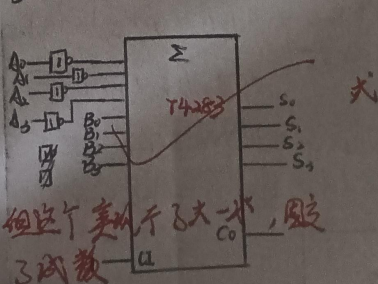
$$Y_9 = A_0 + \bar{A}_1 + \bar{A}_2 + A_3 = \bar{A}_0 + A_1 + A_2 + \bar{A}_3$$

$$Y_{10} = A_0 + \bar{A}_1 + \bar{A}_2 + A_3 = \bar{A}_0 + A_1 + A_2 + \bar{A}_3$$

3-8



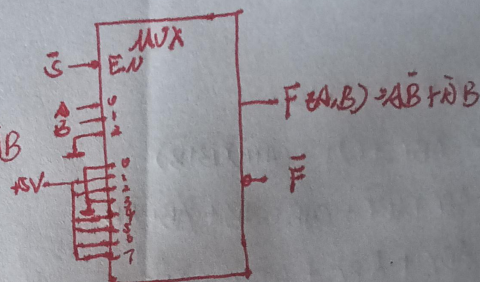
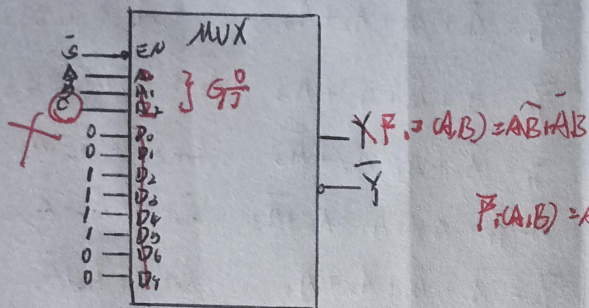
3-10



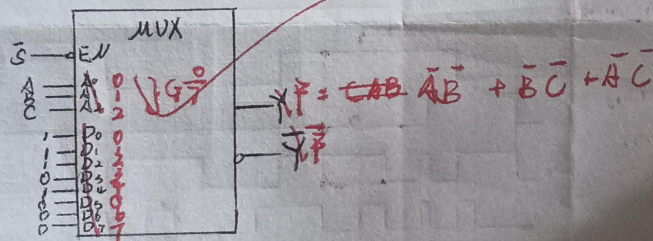
202305011 王立群

3-12

1) $F_1(A, B) = A\bar{B} + \bar{A}B$ 不用，直接查图
 $= A\bar{B}(C + \bar{C}) + \bar{A}B(C + \bar{C})$ 析式
 $= A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}B\bar{C}$ 就是
 $= m_3 + m_4 + m_5 + m_6$ 要化简



2) $F_2(A, B, C) = \bar{A}\bar{B}(C + \bar{C}) + \bar{B}\bar{C}(A + \bar{A}) + \bar{A}\bar{C}(B + \bar{B})$
 $= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$
 $= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$



3) $F_{(4)}(A, B, C, D) = \bar{A}\bar{C}(B + \bar{B})(C + \bar{C}) + \bar{A}\bar{B}D(C + \bar{C}) + \bar{B}\bar{C}D(A + \bar{A})$
 $= (\bar{A}\bar{C} + \bar{A}\bar{B}\bar{C})(C + \bar{C}) + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D$
 $= \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D = \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D$
 $= \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D = \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D$

